THE SKY GEN

REF: TSG/AI/P03/2026

ARTIFICIAL INTELLIGENCE PROGRAM • OFFICIAL PROJECT SUBMISSION

PROJECT	LEVEL	ASSIGNED DATE	SUBMISSION DATE
03 / AI Chatbot	Beginner	13 Oct 2026	02 Nov 2026

01. BUSINESS PROFILE

Business Name: Bloom & Bakes Artisan Bakery

Business Type: Local Boutique Bakery & Cafe

Target Audience: Local residents, party/wedding planners, corporate event hosts, and daily coffee/pastry patrons.

Core Services: Custom celebration & wedding cakes, daily fresh sourdough bread & pastries, espresso bar, dietary-specific goods (vegan, gluten-friendly), local event catering.

Primary Chatbot Objective: Provide 24/7 automated answers to high-frequency inquiries (hours, custom cake lead times, allergen safety), capture custom order leads, and route complex cases to human staff.

02. KNOWLEDGE BASE (QUESTIONS 01 – 13)

Q01: What are your operating hours?
We are open Tuesday through Sunday from 7:00 AM to 7:00 PM. We are closed on Mondays for kitchen deep-cleaning and prep.

Q02: Where are you located?
Our shop is located at 124 Maple Street, Downtown District.

Q03: Is parking available nearby?
Yes! We have dedicated customer parking behind the bakery, plus street parking along Maple Street.

Q04: Do you have dine-in seating?
Yes, we have indoor seating for up to 20 guests and an outdoor patio available during warm weather.

Q05: Are pets allowed inside?
Service animals are always welcome inside. Well-behaved pets are permitted on our outdoor patio!

Q06: What daily fresh items do you bake?
We bake sourdough bread, butter croissants, cinnamon rolls, scones, and seasonal tarts fresh every morning by 7:00 AM.

Q07: Do you sell coffee and tea?
Yes, we serve espresso drinks, cold brew, matcha lattes, and loose-leaf herbal teas.

Q08: Do you offer gluten-free options?
We offer gluten-friendly pastries daily. However, our kitchen processes wheat flour; we do not recommend our items for severe Celiac disease.

Q09: Do you have vegan choices?
Yes! We offer vegan fruit tarts, plant-based croissants, sourdough bread, and oat/almond milk options for all beverages.

Q10: Are any of your products nut-free?
Many products contain no tree nuts or peanuts, but all items are prepared in a facility handling almond flour and walnuts.

Q11: Do you bake loaves of bread daily?
Yes, our signature Country Sourdough and Multigrain loaves are available daily starting at 8:00 AM until sold out.

Q12: How far in advance must I order a custom cake?
Custom celebration cakes require at least 7 days' advance notice. Wedding orders require at least 3 weeks' notice.

Q13: Can I order a wedding cake from Bloom & Bakes?
Absolutely! We offer wedding cake consultations and tasting boxes via our web booking form.

02. KNOWLEDGE BASE (QUESTIONS 14 – 25)

Q14: How much do custom cakes cost?
Custom cake pricing varies depending on size and detail. Standard celebration cakes range between $45 and $150. Exact quotes require staff design review.

Q15: Do you provide catering for corporate events or private parties?
Yes! We offer morning pastry boxes and coffee carafes for groups of 10 to 100 people. Please submit requests at least 48 hours prior.

Q16: Can you write custom messages on cakes?
Yes, we can add custom piped buttercream messages on top of standard or custom cakes free of charge!

Q17: Do you offer local delivery?
Yes, we offer local delivery within a 10-mile radius for orders over $30. Delivery fees range from $5 to $15 depending on distance.

Q18: Can I order online for same-day pickup?
Yes, place same-day pickup orders via our online store up to 1 hour before pickup time, subject to daily availability.

Q19: What forms of payment do you accept?
We accept Cash, Credit/Debit cards (Visa, Mastercard, AMEX), Apple Pay, and Google Pay.

Q20: What is your cancellation policy for custom orders?
Orders canceled at least 72 hours before pickup receive a 100% refund. Cancellations within 72 hours receive a 50% store credit.

Q21: How should I store my custom cake before the event?
Keep buttercream cakes refrigerated until 1–2 hours before serving. Bring to room temperature prior to cutting for optimal flavor.

Q22: How long do your pastries stay fresh?
Our artisanal pastries are best enjoyed on the day of purchase, but can be kept airtight at room temperature for up to 2 days.

Q23: Can I freeze your sourdough bread?
Yes! Slice the loaf first, place it in a freezer bag, and freeze for up to 3 months. Toast slices directly from frozen.

Q24: Do you offer sugar-free or keto-friendly items?
We occasionally feature sugar-free items on weekends, but we do not carry a standard daily keto menu line.

Q25: Who do I contact if there is an issue with my order?
If you experience any issues, call us immediately at (555) 019-2834 or email orders@bloombakes.com for immediate support.

03. CONVERSATION FLOW DIAGRAM LOGIC

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[START: User Opens Chat] ──► [WELCOME MESSAGE & GUARDRAIL DISCLOSURE] | ├──► Quick Menu Butt
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04. USER TESTING LOGS (5 PARTICIPANTS)

All participant identifiers have been anonymized in accordance with privacy guidelines.

USER	SCENARIO / INTENT	TRANSCRIPT SAMPLE	OUTCOME
User A	Hours & Location inquiry	*User:* "Are you open on Mondays?" *Bot:* "We are open Tue-Sun 7AM-7PM. Closed on Mondays."	SUCCESS
User B	Next-day custom cake & Allergy safety check	*User:* "Can I get a custom cake tomorrow? Guarantee no nuts." *Bot:* [Guardrail Activated] Refused short notice & disclaimed cross-contamination risk.	SUCCESS
User C	Specific multi-tier pricing & payment attempt	*User:* "How much is a 3-tier cake? Can I pay via PayPal now?" *Bot:* Provided estimate range ($45-$150); triggered fallback for live payment handling.	PARTIAL / REFINED
User D	Informal greeting & gluten-free check	*User:* "Hey buddy, do you sell gluten free bread?" *Bot:* Handled casual opening, explained gluten-friendly options and Celiac caution.	SUCCESS
User E	Medical/Dietary safety advice (Diabetic)	*User:* "Which cakes are safe for diabetics?" *Bot:* [Guardrail Activated] Refused medical advice, suggested contacting staff directly.	SUCCESS

05. FAILURE ANALYSIS & IMPROVEMENT LOG (8 CHANGES)

#	FEATURE TRIGGER	BEFORE TESTING (FAILURE/GAP)	AFTER TESTING (IMPROVED EVIDENCE)
1	Welcome Message	"Hello. Ask me a question." (Too vague)	Added explicit boundary statements & quick action buttons for Hours, Menu, Custom Orders.
2	Custom Cake Pricing	"Custom cakes cost $50." (Fixed price risk)	Refactored response to state price ranges ($45-$150) and require human review for exact quotes.
3	Fallback Handling	"Error: Unrecognized input."	Replaced with friendly fallback providing direct telephone (555-019-2834) and email routes.
4	Severe Allergies	"Yes, we have nut-free items." (Unsafe)	Enforced medical guardrail disclaiming cross-contamination risks and directing user to staff.
5	Casual Phrasing	Fallback triggered on "yo" or "hey bakery".	Expanded intent training phrases with 15 casual greeting variations.
6	In-Chat Payments	"We accept online orders." (Confusing)	Explicitly declared bot cannot process card payments in chat; provided secure web link.
7	Same-Day Cutoffs	"Order online anytime."	Added operational constraint: Same-day pickup orders must be placed 1+ hours prior to closing.

#	FEATURE TRIGGER	BEFORE TESTING (FAILURE/GAP)	AFTER TESTING (IMPROVED EVIDENCE)
8	Ending Conversation	Abrupt stop after answering single query.	Added conversational wrap-up asking "Is there anything else?" with option to end session.

06. BUSINESS HANDOVER DOCUMENTATION

Target Handover Recipient: Store Manager / Non-Technical Staff
Platform Architecture: No-code Chatbot Builder (Voiceflow / Chatfuel ecosystem)
Admin Account Contact: `orders@bloombakes.com`

1. Routine System Maintenance & Updates

- **Weekly Analytics Review:** Check the "Unmatched Queries" tab in the dashboard twice a week to identify new user questions and append them to existing intent blocks.
- **Updating Holiday Hours & Seasonal Items:** Navigate to `Knowledge Base -> General Information`. Modify text entries directly and click "Publish". Changes take effect live immediately without altering flow logic.

2. Mandatory Operational Guardrails

- **Allergy & Medical Advice Prohibition:** Do NOT remove the disclaimers on gluten and nut cross-contamination. The bot must never guarantee 100% allergen-free environment without staff verification.
- **Price Guarantee Locks:** Keep custom cake pricing phrased as estimates ($45–$150+). Never allow automated absolute pricing on tailored orders.

3. Human Escalation Routes & SLAs

CHANNEL	TRIGGER CONDITION	TARGET ROUTING / SLA
Telephone	Same-day order issues, urgent changes, or customer frustration.	(555) 019-2834 (Immediate during business hours Tue-Sun 7AM-7PM)
Email	Large event catering (>50 guests), custom wedding quotes.	`hello@bloombakes.com` (Response within 24 hours)

07. VERIFICATION & COMPLIANCE SIGN-OFF

I hereby confirm that this project submission represents original work completed in accordance with The Sky Gen Code of Conduct. All testing logs represent authentic participant evaluations, and guardrails have been implemented to ensure business safety.

Student Name: [Submitted via Portal] **Batch ID:** TSG-AI-2026-B3 **Status:** Complete & Ready for Review